

LESSON 3-1 · SESSION 2 · PRACTICE · FORM A

Ratios and Tape Diagrams

Learning target – I can use a tape diagram to share a total according to a given ratio. · 6.RP.A.1

REMEMBER

- A **tape diagram** uses equal boxes. The ratio tells how many boxes each group gets.
- **One box = total ÷ total parts.** Multiply one box by each term.
- If you know one group, divide it by its boxes to find one box.

WORKED EXAMPLE

Ana and Ben share 28 stickers in the ratio 3 : 4. Find each share.

Step 1 Total parts: $3 + 4 = 7$.

Step 2 One box: $28 \div 7 = 4$.

Step 3 Ana $3 \times 4 = 12$; Ben $4 \times 4 = 16$. Check: $12 + 16 = 28$.

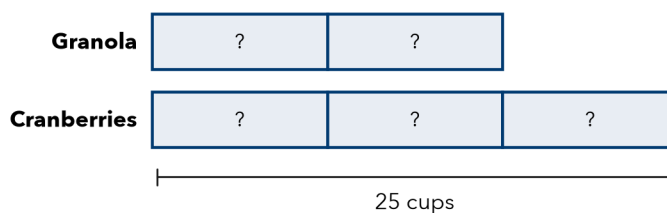
Answer Ana 12, Ben 16

WORDS tape diagram (*diagrama de cinta*) · unit (*unidad*) · total parts (*partes totales*)

PRACTICE

show your work · label units

1. **Trail Mix.** A recipe uses 2 cups of granola for every 3 cups of cranberries. A chef makes a 25-cup batch.



- a. Total parts: _____
- b. Value of one box: _____ cups
- c. Granola and cranberries: _____
2. **Smoothie.** A smoothie uses 3 strawberries for every 2 bananas. A pitcher uses 20 fruits. How many bananas?
3. **Mural Paint.** Art club mixes 4 cups blue with 1 cup white. They need 35 cups in all. How much blue?

Hint Total parts: $3 + 2$. Divide 20 by it.

Show your work.

Answer _____ bananas

Show your work.

Answer _____ cups

4. **Fish Tank.** A tank has goldfish and guppies in the ratio 5 to 3. There are 32 fish in all. How many goldfish?
5. **Camp Counselors.** A camp has 1 counselor for every 6 campers. There are 42 campers. How many counselors are needed?

Hint One box = $32 \div 8$, not $32 \div 5$.

Show your work.

Answer _____ goldfish

Show your work.

Answer _____ counselors

6. **Splitting Money.** Two cousins share \$45 in the ratio 4 : 5.

Hint Add the terms. Divide \$45 by that total.

- a. Cousin 1: \$ ____ b. Cousin 2: \$ ____

7. **Find the Total.** A trail mix uses 2 parts raisins to 5 parts nuts. The recipe uses 15 ounces of nuts.

Hint Nuts are 5 boxes. Find one box first.

- a. How many ounces is one part? ____ oz
b. How many ounces does the recipe make in all? ____ oz

Show your work.

8. **Garden Beds.** A gardener plants beans and corn in the ratio 5 : 3 with 96 seeds in all.

- a. Bean seeds and corn seeds: ____
b. How many more bean seeds than corn seeds? ____
c. Next year she wants equal amounts. What ratio? ____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

9. **Sofia's Work.** A mix uses 2 cups sugar and 5 cups flour, 28 cups in all. Sofia divided 28 by 2 and got 14 cups of sugar. Explain her mistake and find the sugar.

TEST PRACTICE

choose the best answer

10. A ratio of 2 : 3 shares a total of 30. One box is worth –

- A 5 B 6 C 10 D 15

11. A ratio of 1 : 4 shares 25 cookies. The larger share is –

- A 4 B 5 C 20 D 24

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

24 M	12 Q	8 I	4 Z	21 A	7 R	28 A	5 D	20 G	252 X
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#1b	#2	#3	#4	#5	#7b	#8b
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CHECK YOURSELF

- ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____